

Okay I would like to preface by saying this doesn't work LMAO. It kind of does?? It changes colors?? It kind of fades?? I spent a lot of time trying to debug this, but I'm happy to just have it do something tbh.

Here was my thinking:

Each color can be in 4 different states, increasing in pwm value, decreasing in pwm value, held high, or held low. The sample code originally faded one color repeatedly, so I wanted to just apply that same pwm to the three colors (obviously staggered). I got rid of the time to transition since the original code just had a $\wedge$ wave pattern, but this one needed to be held high or low (so it looked like $_/-_$. So then I did the fade pattern (which went high, dec, low, inc) for each color. There was an additional timer for the hold time (since it's double the amount of time it takes for a color to increase or decrease).

The initial state determined where it started in the cycle. So red started halfway through its hold high state, green started without any special defaults (since it needed to start at INC), and blue started at hold low. Then it was just supposed to start counting, with all of the cycles being staggered, but somehow that didn't work. It's probably in how I count the start intervals and states, but I'm kind of done with it right now.

